GRADE 7 MATHEMATICS — TEST PREPARATION

Topics: LCM • GCF • Prime Numbers • Prime Factors • Multiples • BEDMAS • Square Roots • Word Problems

Name: _____
Date: _____

PART A — MULTIPLES AND FACTORS

1. Write the first six multiples of 7.

_____

2. Write the first six multiples of 9.

_____

3. List all the factors of 36.

_____

4. List all the factors of 48.

_____

5. Circle the prime numbers.

11 15 17 21 23 27 31 35

6. Is 37 prime? Explain how you know.

_____
_____

PART B — PRIME FACTORIZATION

7. Write the prime factorization of 36.

_____

8. Write the prime factorization of 60.

_____

9. Write the prime factorization of 72.

10. Write the prime factorization of 90.

PART C — GCF with prime factorization method. Students please check when the numbers are big its hard to find the factors. Then use prime factorization method.

E.g. **Example: Find the GCF of 24 and 36**

First, write the prime factors:

24 = 2 × 2 × 2 × 3

36 = 2 × 2 × 3 × 3

2 x 2 x 3= 12

11. Find the GCF of 24 and 36.

Answer: _____

12. Find the GCF of 42 and 56.

Answer: _____

13. Find the GCF of 45 and 75.

Answer: _____

14. Find the GCF of 48 and 64.

Answer: _____

PART D — LCM

15. Find the LCM of 4 and 6.

Answer: _____

16. Find the LCM of 8 and 12.

Answer: _____

17. Find the LCM of 9 and 15.

Answer: _____

18. Find the LCM of 12 and 18.

Answer: _____

PART E — BEDMAS

19. Solve.

7 + 4 × 6 = _____

20. Solve.

24 ÷ 6 + 8 × 2 = _____

21. Solve.

(15 − 7) × 3 = _____

22. Solve.

30 − 18 ÷ 3 + 4 = _____

23. Solve.

(20 + 4) ÷ 6 + 7 = _____

24. Solve.

5^2 + 12 ÷ 3 = _____

25. Solve.

40 − (6 + 4) × 3 = _____

PART F — SQUARE ROOTS

26. Find √25.

Answer: _____

27. Find √49.

Answer: _____

28. Find √81.

Answer: _____

29. Find √100.

Answer: _____

30. Find √169.

Answer: _____

PART G — WORD PROBLEMS

31. Two school bells ring at different intervals. One rings every 6 minutes and the other rings every 8 minutes. If they ring together at 9:00 a.m., after how many minutes will they ring together again?

Show your work.

Answer: _____

32. A teacher has 36 pencils and 48 erasers. She wants to make identical student packages with no items left over. What is the greatest number of packages she can make?

Show your work.

Answer: _____

33. A sports club has 24 soccer balls and 36 basketballs. They want to arrange the balls into equal groups with the same number of soccer balls and basketballs in each group. What is the greatest number of groups they can make?

Show your work.

Answer: _____

34. A student solves this expression:

18 + 12 ÷ 3 × 2

She gets 20.

Is her answer correct? Solve the expression using BEDMAS and explain.

Answer: _____

35. A square garden has an area of 144 m². What is the length of one side of the garden?

Show your work.

Answer: _____

36. A teacher has 5 boxes. Each box contains 8 packs of pencils. Each pack contains 6 pencils. How many pencils are there altogether?

Write an expression and solve it.

Expression: _____

Answer: _____ pencils

CHALLENGE QUESTIONS

37. Find the smallest number that is a multiple of both 15 and 20.

Answer: _____

38. A number has the prime factorization:

$2 \times 2 \times 3 \times 5$

What is the number?

Answer: _____

39. Use the numbers 2, 3, 4, and 6 to create a BEDMAS expression that equals 14.

Expression: _____

40. A school has two activity groups. Group A meets every 12 days and Group B meets every 18 days. If both groups meet today, after how many days will they meet on the same day again?

Show your work.

Answer: _____

GRADE 7 MATHEMATICS — ANSWER KEY

PART A — MULTIPLES AND FACTORS

1. First six multiples of 7:
7, 14, 21, 28, 35, 42

2. First six multiples of 9:
9, 18, 27, 36, 45, 54

3. Factors of 36:
1, 2, 3, 4, 6, 9, 12, 18, 36

4. Factors of 48:
1, 2, 3, 4, 6, 8, 12, 16, 24, 48

5. Prime numbers:
11, 17, 23, 31

6. Is 37 prime?
Yes. It has exactly two factors: **1 and 37**.

PART B — PRIME FACTORIZATION

7. 36 = **2 × 2 × 3 × 3**

8. 60 = **2 × 2 × 3 × 5**

9. 72 = **2 × 2 × 2 × 3 × 3**

10. 90 = **2 × 3 × 3 × 5**

PART C — GCF

11. GCF of 24 and 36 = **12**

12. GCF of 42 and 56 = **14**

13. GCF of 45 and 75 = **15**

14. GCF of 48 and 64 = **16**

PART D — LCM

15. LCM of 4 and 6 = **12**

16. LCM of 8 and 12 = **24**

17. LCM of 9 and 15 = **45**

18. LCM of 12 and 18 = **36**

PART E — BEDMAS

19. 7 + 4 × 6 = **31**

20. 24 ÷ 6 + 8 × 2 = **20**

21. (15 − 7) × 3 = **24**

22. 30 − 18 ÷ 3 + 4 = **28**

23. (20 + 4) ÷ 6 + 7 = **11**

24. 5^2 + 12 ÷ 3 = **29**

25. 40 − (6 + 4) × 3 = **10**

PART F — SQUARE ROOTS

26. √25 = **5**

27. √49 = **7**

28. √81 = **9**

29. √100 = **10**

30. √169 = **13**

PART G — WORD PROBLEMS

31. 24 minutes
LCM of 6 and 8 = **24**
They ring together again at **9:24 a.m.**

32. 12 packages
GCF of 36 and 48 = **12**

33. 12 groups
GCF of 24 and 36 = **12**

34. No, 20 is incorrect.
18 + 12 ÷ 3 × 2
= 18 + 4 × 2
= 18 + 8
= **26**

35. 12 m
√144 = **12**

36. 240 pencils
5 × 8 × 6 = **240**

CHALLENGE QUESTIONS

37. 60
LCM of 15 and 20 = **60**

38. 60
2 × 2 × 3 × 5 = **60**

39. One possible answer:
6 × 3 − 4 = 14
(Uses 6, 3 and 4, but **does not use 2**, so this question as written needs correction if all four numbers must be used.)

A corrected expression using all four numbers is:
6 + 4 × 3 − 2 = 16, not 14.

So **Question 39 should be revised** rather than marked with an answer.

40. 36 days
LCM of 12 and 18 = **36**.